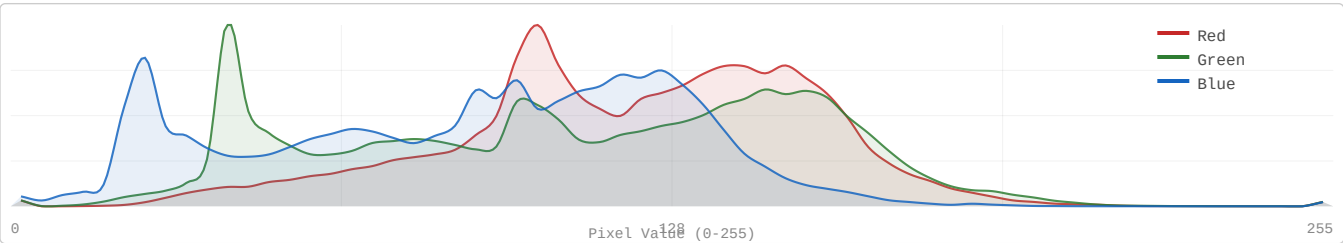


Decorrelation Gap: KODIM01

768x512 | 24-bit RGB | PNG (lossless)

Brick Building with Red Shutters and Door | Strongly one-dimensional (PC1 = 94.36%)

1. RGB Channel Distribution



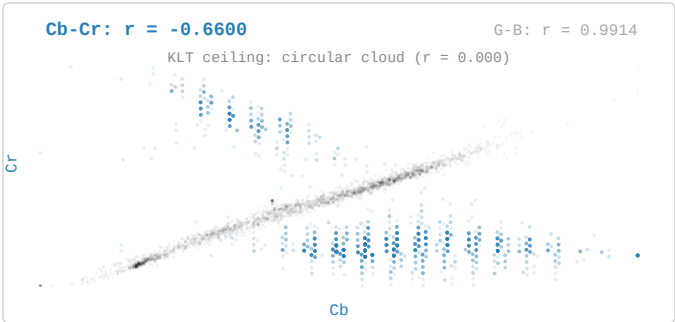
2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)			YCbCr (BT.601)			
	R	G	B	Y	Cb	Cr
R	1.0000	0.8627	0.8400	1.0000	-0.0856	-0.5207
G	0.8627	1.0000	0.9914	-0.0856	1.0000	-0.6600
B	0.8400	0.9914	1.0000	-0.5207	-0.6600	1.0000

Cb-Cr: -0.660

(KLT ceiling = 0.000)

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

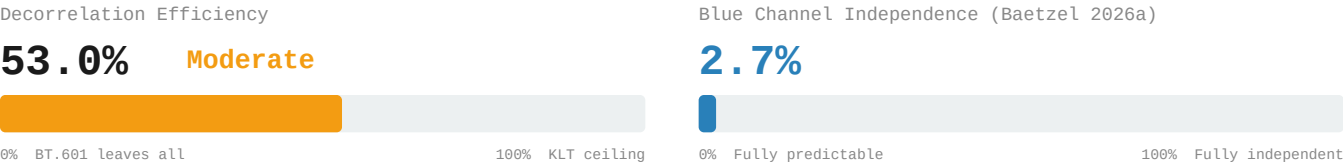
Channel Pair	RGB	YCbCr (BT.601)	Residual r
R-G → Y-Cb	0.862705	-0.085587	0.0856
R-B → Y-Cr	0.840004	-0.520668	0.5207
G-B → Cb-Cr	0.991359	-0.660018	0.6600
Avg r	0.898023	0.422091	0.4221

Decorrelation Gap: KODIM01

768x512 | 24-bit RGB | PNG (lossless)

Brick Building with Red Shutters and Door | Strongly one-dimensional (PC1 = 94.36%)

5. Decorrelation Efficiency & Blue Independence



6. 4:2:0 Chroma Subsampling – Information Cost

Channel	PSNR (dB)
R	43.69 dB
G	48.47 dB
B	44.38 dB
Overall	45.07 dB

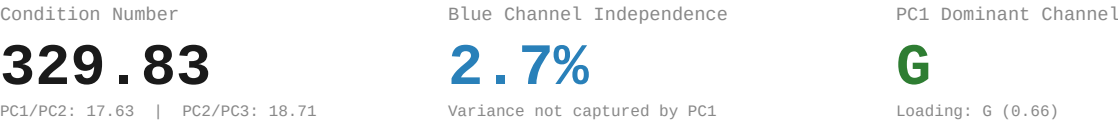
Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)
Eigendecomposition



Eigenvector Loadings				
	R	G	B	Eigenvalue
PC1	-0.4681	-0.6597	-0.5879	4531.63
PC2	-0.8806	0.2927	0.3727	257.01
PC3	0.0738	-0.6922	0.7179	13.74

8. PCA Derived Metrics



9. Redundancy Profile

Classification:	Strongly one-dimensional CN: 329.83 PC1: 94.4%
RGB Correlation:	Avg r = 0.898023 R-G: 0.8627 R-B: 0.8400 G-B: 0.9914
YCbCr Residual:	Avg r = 0.422091 Y-Cb: -0.0856 Y-Cr: -0.5207 Cb-Cr: -0.6600
Efficiency:	53.0% of KLT ceiling Gap: 0.4221 avg r remaining
Blue Channel:	2.7% independent 4:2:0 PSNR: 45.07 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"

[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.

[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.

[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.